

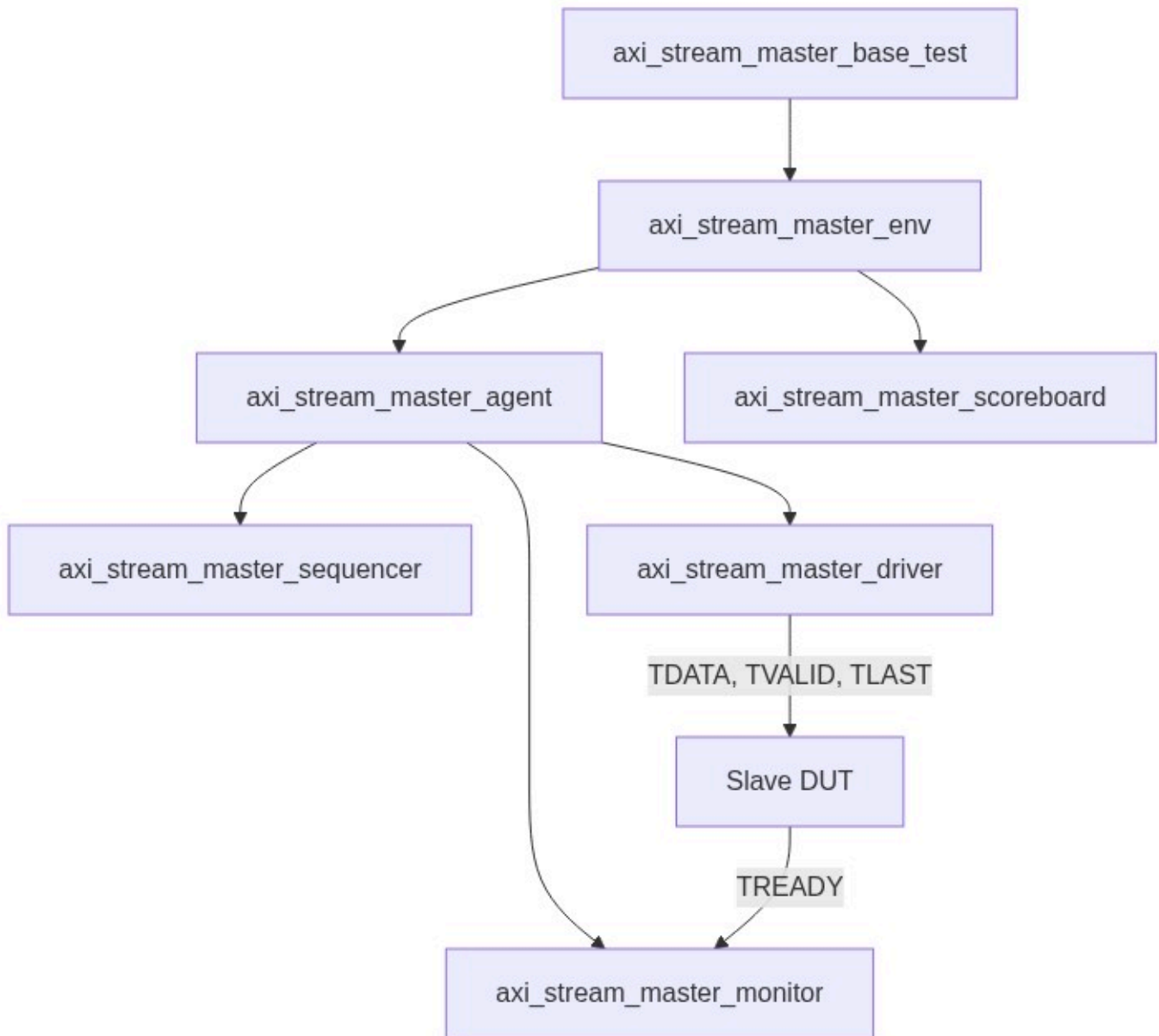
Verification Plan Report

Project: AXI-Stream Exhaustive Verified

Version: 3.3.1

Verification Role: Master

1. UVM Environment Architecture



2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_MS_01	Handshake (Stability)	Data Stability	Master shall keep TDATA stable while Slave DUT holds TREADY LOW.

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REQ_MS_02	Handshake (Independence)	Valid Latency	Master must drive TVALID without waiting for Slave TREADY.
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3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MS_01	Simple Handshake: Observe Slave DUT latency and verify data stability.	REQ_MS_01
TC_MS_02	Reset Recovery: Master VIP drives TVALID cycle 1 after reset. Verify Slave reaction.	REQ_MS_02

4. Functional Coverage Groups

Group Name	Description
cg_handshake	Cross of TVALID x TREADY.

5. Interface Signals

Signal Name	Direction	Width	Description
TDATA	Output	TD_W	Data.
TVALID	Output	1	Valid.
TREADY	Input	1	Ready (Observed).

6. Verification Environment Structure

`verif/axis_exhaustive_v3_3/`

7. SystemVerilog Implementation

`axi_stream_master_seq_item.sv` (Blueprint)

```
COMPONENT: axi_stream_master_seq_item
EXTENDS: uvm_sequence_item
MEMBERS:
- [rand logic [TD_W-1:0]] tdata
- [rand logic [TK_W-1:0]] tkeep
- [rand logic] tlast
LOGIC:
- Basic randomization constraints for AXI-Stream beats.
```

`axi_stream_master_driver.sv` (Blueprint)

```
COMPONENT: axi_stream_master_driver
```

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EXTENDS: uvm_driver

LOGIC:

- Action: Drive TVALID/TDATA.
- Reaction: Observe Slave DUT TREADY for handshake.

axi_stream_master_monitor.sv (Blueprint)

COMPONENT: axi_stream_master_monitor

EXTENDS: uvm_monitor

LOGIC:

- Observe: Sample TVALID && TREADY confluence on ACLK.

axi_stream_master_scoreboard.sv (Blueprint)

COMPONENT: axi_stream_master_scoreboard

EXTENDS: uvm_scoreboard

LOGIC:

- Compare: Match observed transfers against expected reference model.

axi_stream_master_base_seq.sv (Blueprint)

COMPONENT: axi_stream_master_base_seq

EXTENDS: uvm_sequence

LOGIC:

- Control: Orchestrate start_item/finish_item handshake with sequencer.

axi_stream_master_env.sv (Blueprint)

COMPONENT: axi_stream_master_env

EXTENDS: uvm_env

LOGIC:

- Hierarchy: Connect monitor to scoreboard analysis port.